Photoelectric effect

The emission and absorption of light was an early subject for investigation by German physicist Max Planck. As Planck attempted to formulate a theory to explain the spectral distribution of emitted light based on a classical wave model, he ran into considerable difficulty. Classical theory (Rayleigh-Jeans law) predicted that the amount of light emitted from a hot, glowing body (a black body) would increase dramatically as the wavelength decreased, whereas experiment showed that it approached zero. This discrepancy became known as the ultraviolet catastrophe. Experimental data for the radiation of light by a black body showed a distribution having a peak at a wavelength inversely proportional to temperature (Wien's law). The classical prediction for the distribution had no maximum at all. In order to reconcile theory with laboratory results, Planck was forced to develop a new model for light called the quantum model. In this model, light is emitted in small, discrete bundles or quanta.

Theoretical background

Planck's quantum theory

By the late 1800's many physicists thought that they had explained all the main principles of the universe and discovered all the natural laws. But as scientists continued working, inconsistencies that couldn't easily be explained began showing up in some areas of study.

In 1901 Planck published his law of radiation. In it he stated that an oscillator, or any similar physical system, has a discrete set of possible energy values or levels; energies between these values never occur. Planck went on to state that the emission and absorption of radiation is associated with the transitions or jumps between two energy levels. The energy lost or gained by the oscillator is emitted or absorbed as a quantum of radiant energy, the magnitude of which is expressed by the equation:

$$E = hf \tag{1}$$

where E equals the radiant energy, f is the frequency of the radiation, and h is a fundamental constant of nature—we now know it as Planck's constant. Planck's constant was found to have significance beyond relating the frequency and energy of light; it has become a cornerstone of the quantum mechanical view of the subatomic world. In 1918, Planck was awarded a Nobel prize for introducing the quantum of light.

The Photoelectric effect

In photoelectric emission, light strikes a material, causing electrons to be emitted. The classical wave model predicted that as the intensity of incident light was increased, the amplitude and thus the energy of the wave would increase. This would then cause more energetic photoelectrons to be emitted. The new quantum model, however, predicted that higher frequency light would produce higher energy photoelectrons, independent of intensity, while increased intensity would only increase the number of electrons emitted (or photoelectric current). In the early 1900s several investigators found that the kinetic energy of the photoelectrons was dependent on the wavelength, or frequency, and independent of intensity, while the magnitude of the photoelectric current, or number of electrons was dependent on the intensity as predicted by the quantum model. Einstein applied Planck's theory and explained the photoelectric effect in terms of the quantum model using his famous equation for which he received the Nobel prize in 1921:

$$E = hf = \mathrm{KE}_{\max} + W_0 \tag{2}$$

where $\mathrm{KE}_{\max}$ is the maximum kinetic energy of the emitted photoelectrons, and W_0 is the 'work function'—the extra energy needed to remove the electrons from the surface of the material. E is the energy supplied by the quantum of light known as a photon.

The h/e experiment

A light photon with energy hf is incident upon an electron in a cathode of a vacuum tube. The electron uses a minimum W_0 of its energy to escape the cathode, leaving it with a maximum energy of $\mathrm{KE}_{\max}$ in the form of kinetic energy. Normally the emitted electrons reach the anode of the tube, and can be measured as a photoelectric current. However, by applying a reverse potential V between the anode and the cathode, the electrons are prevented from reaching the anode, and the photoelectric current can be stopped. $\mathrm{KE}_{\max}$ can be determined my measuring the minimum reverse potential needed to stop the photoelectrons and reduce the photoelectric current to zero (although note that we will use a slightly different method). Relating kinetic energy to stopping potential gives the equation:

$$\mathrm{KE}_{\max} = eV \tag{3}$$

Therefore, using Einstein's equation,

$$hf = eV + W_0 \tag{4}$$

When solved for V, the equation becomes:

$$V = (h/e)f - (W_0/e) \tag{5}$$

Thus if we plot V as a function of f for different frequencies of light, we expect to find a linear relationship between the frequency of light and the stopping potential V. This relationship will have a V intercept equal to $-W_0/e$ and a slope of h/e. If we combine our experimental determination of the ratio h/e with the accepted value for e, 1.602×10^{-19} C, we can determine Planck's constant, h.

Experimental setup

Set up the equipment as shown in Fig 1. Turn on the mercury lamp. It may take a few seconds to fully light up. The light from the lamp should pass through the Lens/Grating assembly, which will produce different colors of light at different angles from the center line.

Focus the light from the Mercury Vapor Light Source onto the slot in the white reflective mask on the h /e Apparatus. Tilt the light shield of the Apparatus out of the way to reveal the curved white photodiode mask inside the Apparatus (see Fig 2). When the arm connecting the h/e Apparatus and lamp is straight, you should see a bright band with sharp edges from the lamp light incident on the white mask; if necessary, loosen the screws to slide the Lens/Grating assembly forward and back on its support rods until you achieve a sharp image, and then re-secure the Lens/ Grating assembly by tightening the thumbscrew.

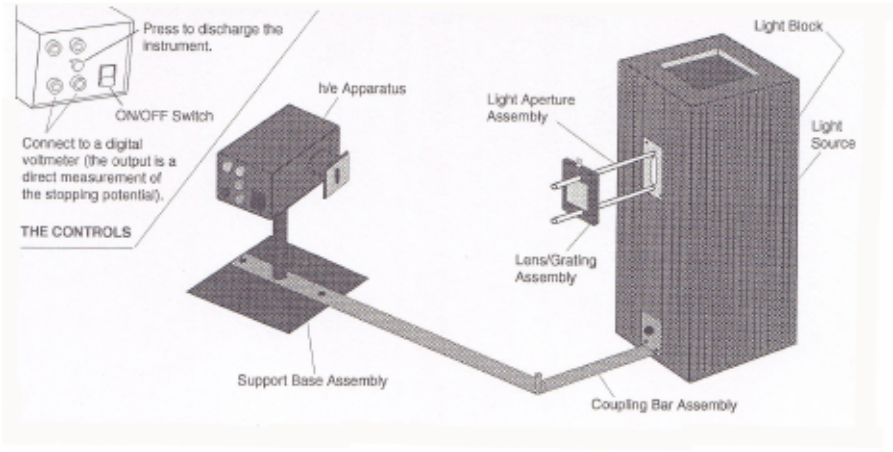


Figure 1: Photoelectric effect experimental setup

Also rotate the h/e Apparatus from side to side and identify the small black square on the curved white mask inside. This square is the hole for the photodiode, and the band of light light must be aligned with it for *every measurement* that you make.

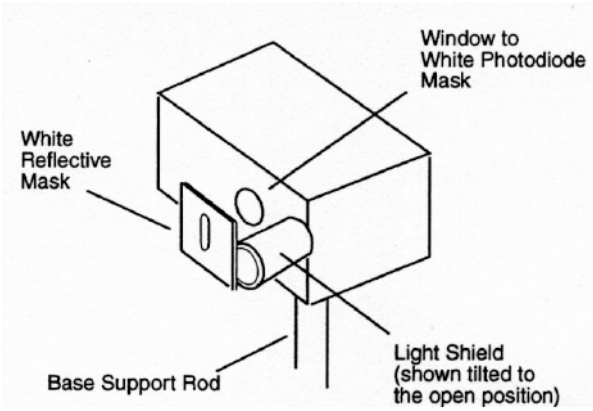


Figure 2: Detail of photodiode housing

Check the polarity of the leads from your digital voltmeter (DVM), and connect them to the OUTPUT terminals of the same polarity on the h/e Apparatus.

The h/e Apparatus includes two filters: one Green, and one Yellow. The filter frames have magnetic strips and mount to the outside of the White Reflective mask of the h /e Apparatus. Use the green and yellow filters when you're using the green and yellow spectral lines. These filters limit higher frequencies of light from entering the h /e Apparatus. This prevents ambient room light from interfering with the lower energy yellow and green light and masking the true results. It also blocks the higher frequency ultraviolet light from the higher order spectra which may overlap with lower orders of yellow and green.

Instead of applying a reverse voltage manually to find the stopping potential (when the current stops), we let the freed electrons accumulate across the face of a capacitor. This generates a reverse potential. When enough charges have accumulated, the voltage across the capacitor prevents charges from moving across the gap and the voltage stabilizes. A button on the side of the detector can discharge the capacitor, allowing charges to flow again.

Procedure

1. Gently swing the arm of the h/e Apparatus toward you until you see some bluish lines from the

lamp appear on the outer reflective mask. Then open the light shield as in Fig 2, and swivel the head of the apparatus so that only the first ultraviolet line passes through the outer mask and lands on the black square of the photodiode aperture in the inner mask. Then close the light shield.

2. With the first ultraviolet line still passing through to the photodiode, measure the stopping potential with the multimeter and record that measurement in a table. Before reading the multimeter, press the "PUSH TO ZERO"button on the side panel of the h/e Apparatus to discharge any accumulated potential in the unit's electronics. This will assure the Apparatus records only the potential of the light you are measuring. *Note that the output voltage will drift slowly, so determine how you will decide when and how to record a specific voltage.*

3. Continue this process with the rest of the lines: you should see five colors in at least two orders of the mercury light spectrum (see Fig 3). For each line, gently swing the arm and swiveling the h/e Apparatus with the light shield open to ensure that only the line of interest passes through the mask and lands on the photodiode. When measuring the yellow and green spectral lines, be sure to place the yellow or green colored filters on the Reflective Mask of the h/e Apparatus to block emission from higher orders from contaminating your measurement.

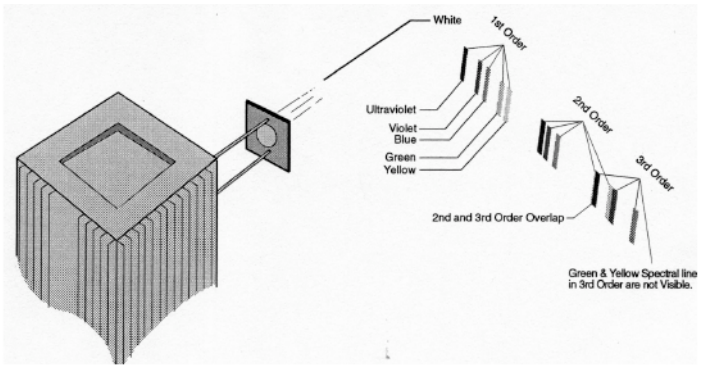


Figure 3: The Mercury lamp spectrum

4. Find uncertainties in your stopping potentials by repeating the measurement of each several times. Consider also the intrinsic uncertainty due to the multimeter itself.

5. Once you have completed measuring the stopping potential for all the emission lines in the first order, move to the second order and repeat the process for the same colors. Record your results.

Analysis

1. Determine the wavelength and frequency of each spectral line. Plot a graph of the stopping potential vs. frequency—note that the wavelengths of each emission line of mercury is gven in the table below. Make separate plots for your first order measurements and your second order measurements.

2. Determine the slope and y-intercept. Interpret the results in terms of the h/e ratio and the W_0/e ratio. Calculate h and W_0 and their uncertainties.

3. Describe potential sources of random and systematic uncertainty in your measurements and how they may have affected your final result. Also describe any changes to your methods or technique that could reduce these uncertainties.

4. In your discussion, report your values and discuss your results with an interpretation based on a quantum model of light. Be sure to compare your measurement of h to the accepted value, as well as to compare the measurements you made via each of the two orders of emission lines.

The mercury spectrum

Color	Wavelength (nm)
Yellow	578.0
Green	546.1
Blue	435.8
Violet	404.7
Ultraviolet	365.5